NOT MEASUREMENT SENSITIVE

A-A-52422
January 12, 1993
SUPERSEDING
(see 5.5)

COMMERCIAL ITEM DESCRIPTION

# BELTS, V: INDUSTRIAL, DRIVE, NARROW TYPE

The General Services Administration has authorzed the use of this commercial item description (CID) as a replacement for MIL-B-52919A, MS39277A, MS39278A, and MS39279A, whIch are canceled.

1.0 Abstract. This CID covers endless, narrow, sIngle and multiple-drive V-belts of three different cross-sectional designations and of various lengths within each designation (see 2.4.1 and 2.4.2). The belts are used in transmitting power in industrial applications.

1.1 CID coverage. This CID does not include all varieties of belts as indicated by the title (see 2.4.1} .

2.0 Salient characteristics.

2.1 Description. The V-belts shall be endless loops having a trapezoidal cross section, measured in inches and millimeters (mm) .

2.2 Material. Material shall be as specified herein. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.7).

2.2.1 Cord. Cords shall be natural or synchetic fibers twisted into a cord and treated for low growth and adhesion to the supporting polymer.

2.2.2 Fabric. Fabric shall be woven or knit natural or synthetic fibers Impregnated with rubber.

2.2.3 Rubber. Rubber shall be either natural or a synthetic polymer compound.

2.3 Construction. The belts shall consist of tension members (inserts) and a compression member. The members shall be tightly bonded with rubber, forming an integral unit. A cover may be used (see 2.3.3).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

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2.3.1 Tension members. Tension members shall be either cord or fabric and shall be arranged in a symmetrical manner about the pitchline of the belt and imbedded in the belt structure.

2.3.2 Compression member. The compression member shall form the narrower trapezoided portion of the belt in the section below the tension members and shall support the belt in the pulleys. The base that forms the inner circumference of the belt may be plain, cogged, or notched, The compression member shall be rubber and may contain cords or fabrics.

2.3.3 Cover. The cover, if used, shall consist of one or more plies of stretchable fabric or cord, frictioned or impregnated for adhesion

2.4 Dimensions.

2.4.1 Cross-sectional designation. The nominal cross-sectional designations in inches and millimeters shall be as shown below, and the belt angle shall be determined by the rideout and bottom clearance.

| | |
|---|---|
| 3V/9N | 3V - 0.38-inch wide by 0.31-inch thick and<br>9N - 9 mm wide by 8 mm thick. |
| 5V/15N | 5V - 0.62-inch wide by 0.53-inch thick and<br>15N - 15 mm wide by 13 mm thick. |
| 8V/25N | 8V - 1.00-inch wide by 0.91-inch thick and<br>25N - 25 mm wide by 23 mm thick. |

2.4.2 Lengths. The standard effective lengths and length tolerance shall be as specified in the engineering standard IP-22 of the rubber manufacturers association.

2.4.3 Rideout and bottom clearance. The position of the top of the belt with respect to the top of the measuring pulley groove (rideout) and bottom clearance between the bottom of the measuring pulley groove and the bottom of the belt shall conform to the tolerances specified in table I.

2.5 Breaking strength and elongation. The belts shall have the minimum breaking strength shown in table II. The elongation of the belts at the minimum specified breaking strength shall not exceed 8 percent.

2.6 Electrical resistivity. When specified (see 5.2), electrical resistance shall not be greater than 6 megohms.

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TABLE I. Details for dimensional checking of V-belts.

| Belt size designation | Pulley O.D. | | Pulley effective outside circumference | | Pulley groove angle | Pulley groove top width | | Pulley groove depth (minimum) | |
|---|---|---|---|---|---|---|---|---|---|
| | ±0.005 inch | ±0.1 mm | inch | mm | ±0.25 degree | ±0.005 inch | ±0.13 mm | inch | mm |
| 3V | 3.820 | - | 12.000 | - | 38 | 0.350 | - | 0.340 | - |
| 9N | - | 95.5 | - | 300 | 38 | - | 8.89 | - | 8.6 |
| 5V | 7.958 | - | 25.000 | - | 38 | 0.600 | - | 0.590 | - |
| 15N | - | 191.0 | - | 600 | 38 | - | 15.24 | - | 15.0 |
| 8V | 15.916 | - | 50.000 | - | 38 | 1.000 | - | 0.990 | - |
| 25N | - | 318.3 | - | 1000 | 38 | - | 25.40 | - | 25.1 |

| Belt size designation | Total tension | | Rideout (maximum) | | Bottom clearance (minimum) | |
|---|---|---|---|---|---|---|
| | pounds | Newtons | inch | mm | inch | mm |
| 3V | 100 | - | 0.10 | - | 0.08 | - |
| 9N | - | 445 | - | 2.5 | - | 2.0 |
| 5V | 225 | - | 0.12 | - | 0.12 | - |
| 15N | - | 1000 | - | 3.0 | - | 3.0 |
| 8V | 500 | - | 0.16 | - | 0.16 | - |
| 25N | - | 2225 | - | 4.1 | - | 4.1 |



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TABLE II. Breaking strength.

| Designation | Breaking strength, minimum | |
|---|---|---|
| | Pounds | Kilograms |
| 3V | 400 | - |
| 9N | - | 181 |
| 5V | 900 | - |
| 15N | - | 408 |
| 8V | 2000 | - |
| 25N | - | 907 |

2.7 Resistance to temperature extremes. The belts shall not be damaged when exposed to a temperature of 158 degrees Fahrenheit (°F), plus or minus 3.6°F, for 12 hours. The belts shall neither crack nor require a greater torque for the applicable cross-sectional type designations than is specified in table III while exposed to temperatures of minus 32°F, plus or minus 3.6°F.

TABLE III. Maximum torque requirements.

| Cross-sectional type designation | Torque required to start | | Torque required to continue | |
|---|---|---|---|---|
| | ft. lb | N m 1/ | ft. lb | N m |
| 3V | 25 | - | 12 | - |
| 9N | - | 34 | - | 16 |
| 5V | 30 | - | 15 | - |
| 15N | - | 41 | - | 20 |
| 8V | 35 | - | 20 | - |
| 25V | - | 47 | - | 27 |

1/ Newton Meter per radian.

2.8 Oil resistance. When specified (see 5.2), the belt thickness shall increase not more than 20 percent or decrease (shrink) not more than 1 percent after exposure to ASTM oil no. 1 and ASTM oil no. 3, as cited in ANSI/ASTM D471, at a temperature of 158°F for 22 hours.

2.9 Age. When specified (see 5.2), the age of the belts shall be not more than 12 months (four quarters) from the time of manufacture to the date of acceptance,

2.10 Hatched sets. When matched sets of belts are required, the individual belts of the matched set shall conform to the dimensional requirements for single belts; however, the maximum length difference between the shortest and longjest belts in any one set shall not exceed the matching limits shown below:



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| Cross-section designation | Matching limits for one set | |
|---|---|---|
| | inch | mm |
| 3V/9N | 0.15-0.30 | 4-10 |
| 5V/15N | 0.15-0.60 | 4-16 |
| 8V/25N | 0.30-0.75 | 6-24 |

2.11 Marking.

2.11.1 Identification. Identification shall be permanent and legible and shall include as a miminum the manufacturer's (CAGE) code, the part identification number (PIN), the industry standard belt number, and the national stock number (NSN) (see 5,2 and 5.3). When age is specified (see 5.2), the marking shall include the quarter and year, or month and year or week and year. Identification marking shall be accomplished without adversely affecting the acceptability and function of the finished products.

2.11.2 Matched sets. Each belt of a matched set shall be marked as specified in 2.11.1. In addition, belts of a matched set shall be tied together and include a tag or label printed with the following notice: "WARNING, THIS IS A MATCHED SET. DO NOT BREAK TIES, EXCEPT AT INSTALLATION".

2.11.3 Special marking. In addition to the applicable marking specified in 2.11.1 and 2.11.2 each belt that is electric resisting (static conducting) shall be marked "SC".

2.12 Workmanship. Workmanship shall be such as to assure a product free of indentation, press lap, flash untrinmed, blister, torn cover or other defects which may affect its performance, serviceability or appearance.

3.0 Quality assurance provisions.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (exammations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.



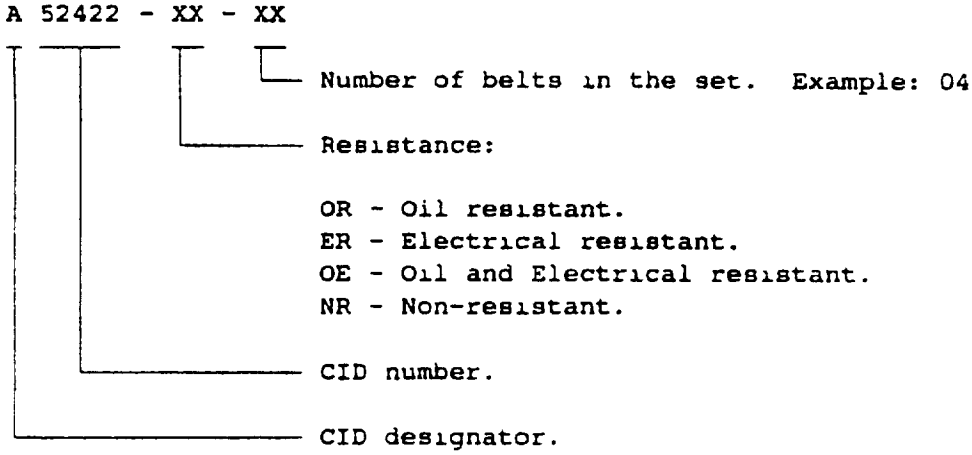
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5 .1.1 Non-Government publications. ANSI/ASTM D471 “Standard Method for Rubber Property, Effect of Liquids" is available from the Society forl Testing and Materials {ASTM), 1916 Race Street, Philadelphia, PA 19103. IP-22 (1991) “Narrow V-Belts and Sheaves” - Engineerng Standard, is available from the Rubber Manufacturers Association, Inc. (RMA), 1400 K Street, N.W. Washington, D.C. 20005.

5.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 5.1).
c. Part identification number (PIN) and the industry standard belt number (see 2.11.1 and 5.3).
d. When belts having electrical resistivity (static conductance) are required (see 2.6).
e. When belts haing oil resistance are required (see 2.8).
f. When belts shall be not more than 12 months old (see 2.9).
g. Selection of applicable level and packaging requirements (see 4).

5.3 Part or identification number (PIN). The PIN’s to be used for the belts acqired to this CID are created as follows:




**NOTE: In addition to the PIN, the industry standard belt number should be specified (see 2.11.1). EXAMPLE 1: 3V1160. 3V indicates cross sectional** designation in inches and 1160 indicates effective belt length in tenths of an inch or 116.0 inches. EXAMPLE 2: 25N12700. 25N indicates cross sectional designation in mm and 12700 indicates effective length in mm. See 2.4.1 and 2.4.2 for details of cross-sectional designation and lengths.

5.4 Metric products.

a. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of ASTM E380, and all other requirements of this CID are met.

b. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable .



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c. The contracting officer has the option of accepting or rejecting the product.

5.5 International standardization. Certain provisions of this CID are the subject of International Standardization Agreement ABCA 63. When amendment, revision, or cancellation of this CID is proposed which could affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliatlon action through international standardization channels including departmental Standardization offices, if required.

5.6 Supersession and cross-reference data. This CID supersedes MIL-B-529l9A, dated 29 June 1981 and MS39277A, MS39278A, and MS39279A, dated 22 June 1981. Belts confirming to this CID are interchangeable/subst~tutable with belts conforming to MIL-B-52919A, MS39277A, MS39278A, and MS39279A.

5.7 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

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